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(1) Research Program/Project/Study Title: Pangisda: Exploring Fishers' Indigenous Knowledge and Physics Concepts	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☒ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☐ Food Research ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:	Roxas City						
		Province:	Capiz						
		Region:	Western Visayas						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☒ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☐ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☒ Basic	☐ Pilot Testing	☐ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☒ Basic	☐ Pilot Testing								
☐ Applied	☐ Tech. Promotion/Commercialization								

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(7) Sector/Commodity

Social Science/ Education/Fisheries/Traditional Knowledge/Culture

(8) Discipline/Field of Specialization

Science Education and Fisheries

(9) Rationale (State the research problem and significance of the project to the current needs of the country. The proposal should justify resource expenditure. A typical justification would include a brief introduction, a general statement concerning the historical basis for R & D and, thrusts being addressed, utilization of the expected output and the impact information generated will have on science, the target users and the country.)

Human mind is the source of all knowledge The process of creating knowledge is impacted by the social environment in which it happens, which includes both cultural and economic factors intellectual norms. The social environment influences the status ascribed to specific knowledge, which is impacted by power relations between those who develop and those who use the information. In the case of scientific knowledge, the scientific revolution ushered in a new era of knowledge generation and verification that many feel is widely regarded as objective, culture-agnostic, and emblematic of the highest form of logical reasoning. Some, however, claim that "scientific knowledge" is actually a Western cultural artifact (Dussel, 2000). Other forms of knowledge are then referred to as 'indigenous knowledge'. Understandings, abilities, and philosophies established by local groups with extensive histories and experiences of engagement with their natural surrounds are referred to as indigenous knowledge (Hiwasaki, et al 2015). However, indigenous knowledge is sometimes ignored or even refuted because it seems to be at odds with scientific knowledge. While Indigenous knowledge sometimes contrasts with scientific knowledge, it can also be complementary and provide supplementary information about the world. Indigenous knowledge, makes frequent use of good scientific ideas, usually experimentally developed, but not always explained in traditional scientific terms.

It is important to conserve indigenous knowledge for future generations. Since it is typically well-suited to address unique requirements in local contexts utilizing locally accessible resources (Das Gupta, 2011; Shaw, Takeuchi, Uy, & Sharma, 2009).

From an educational perspective, the level to which indigenous knowledge is used influences the extent to which science and technology contribute to the quality of life in developing nations such as the Philippines. Thus, there should be a strong degree of emphasis on indigenous knowledge in science and technical education.

(9-A) Project Description (Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other works needed to carry out the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)

In order to ascertain the indigenous and physics knowledge of fishers of Roxas City, Capiz, the following procedures will be employed.

(1) Identification of participants of the study

A convenience sample of 25 as a good number of informants for qualitative research (Moser & Korstjens 2018; Creswell, 1998) to achieve data saturation.

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(2) Interview

In-depth, unstructured, individual interviews will be conducted. At the beginning of the interview, respondents will be asked to talk about their daily activities, practices, and experiences regarding fishing. The interview will be audio-recorded. Thus, the interviewer will listen attentively for personal opinions and information that could be associated with physics concepts and phenomena, asking follow-up questions and requesting further elaboration.

(3) Field Observation

To validate the themes related to physics, observation on the actual fishing practices will be conducted to give the researcher a clearer view on the indigenous knowledge of the respondents. Field observations allow researchers to collect rich and meaningful data from participants in their natural settings (Meriam, 1998).

(4) Data analyses and interpretation

The interview transcripts will be analysed to identify those ideas that could be related to physics concepts. After the identification of different physics concepts, coding of qualitative data into clusters of similar entities will be conducted using Thematic Analysis by Braun and Clarke (2006). As each theme emerge it will be coded, analyzed, grouped, and organized.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

This study aims to ascertain the indigenous and physics knowledge of fishers of Roxas City, Capiz.

Specifically it sought to answer the following questions:

1. What are the indigenous knowledge of fishers' related to fishing ?
2. What are the indigenous practices of fishers?
3. What are the physics concepts behind the indigenous knowledge and practices of fishers?

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of –the-art of current technology / information from which the project proposal will take-off).

Indigenous knowledge

Indigenous Knowledge has become the recognized phrase for non-western people's beliefs and understandings gained from long-term interaction with a region. It is knowledge founded on social, physical, and spiritual understandings that has shaped people's survival and contributed to their sense of place in the world. Indigenous Knowledge is also known as Traditional Ecological Knowledge (TEK), Indigenous People's Knowledge (IPK), and even 'folk wisdom'. Indigenous knowledge is defined as knowledge which is spatially and/or culturally context specific, collective, holistic, and adaptive (Mistry, 2009). According to the UNESCO's programme on Local and Indigenous Knowledge Systems (LINKS), Indigenous knowledge refers to understandings, skills, and philosophies developed by local communities with long histories and experiences of interaction with their natural surroundings.

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A survey was conducted by Raju, et al (2016) to assess the indigenous fishing methods adopted by the fisherman in Lake Kolleru. According to the study, the fishermen have created a diversified variety of fishing equipment and tactics over a lengthy period of time in order to catch a vast assortment of fish species, and economic use of the lake.

Indigenous knowledge and other related studies

Various studies explores the connection and importance of indigenous knowledge in different fields of environmental management (Bardsley & Wiseman, 2016; Lemi, 2019; Gail, 2011), disaster risk reduction (Fang, et al, 2008; Takeuchi & Shaw, 2008, Borkhuu, 2008), education (Handayani,2018; Snively,2001).

Baquetea, Graysona, & Mutimucuio (2016) explored the farming indigenous knowledge of senior citizens towards related to physics concepts. Indigenous fishing practices with Physics applications include Paghikot (net fishing by walking and using a banca), Pagpana (spearfishing by day and night), and Pagbanwit (hook and rod): pag-og-og, pagsibid-sibid, pagtalunton, pagkanuos with a banca, pagkanuos via balakwit, and pagbalakwit. The following physics concepts are present in indigenous fishing practices: force, motion, center of mass, pressure, buoyancy, elasticity, work, momentum, light, and sound.

The major argument of this study is quite parallel to the local work of Besmonte and Mina (2021). This study used ethnographic method through direct field observation and immersion, interviews and describe the physics principles manifested in the fishing practices of the fisherfolks of coastal barangays in Albay. However, since fishing knowledge and practices differ from one place to another, thus this study is worth exploring.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories and variables that indicate relevant relationships to be tested.

Indigenous knowledge and practices often makes use of sound scientific principles, typically empirically derived, but not necessarily explained in conventional scientific terms. It is important to maintain such indigenous knowledge for both cultural and practical reasons, since it is frequently well-suited to meeting unique demands in local contexts using locally accessible resources. In an educational perspective, the science behind, specifically physics in this study incorporated or manifested in indigenous knowledge and principle could have a positive effect on understanding our culture and uplifting scientific literacy in the community. Figure 1 shows the framework of the study.

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(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible)

Indigenous knowledge → Scientific knowledge → Fisheries Education → Scientific Literacy

It is also a projected output of this study to produced coffee table books showcasing the different indigenous fishing knowledge and practices of the fishers in Roxas City and the physics concepts behind.

(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

1. Copyrighted scientific/artistic work
2. Curriculum plan (Inclusion of results in fisheries curriculum or plan, if possible)
3. Public exhibit

(17) Target Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

The results of this study may serve as a baseline data for fisheries educators and curriculum planner for the inclusion of indigenous knowledge in science subjects.

(18) Personnel Requirement

Name	Role (Program,Project, Study Leader)	Field of Specialization

Percent time devoted to the Program/ Project/ Study

Program/Project Leader (in hours):

Program/Project Members (in hours):

(19) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needed the Research Assistant, informant/s, guide)	2 RAs	P22,000.00
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance		10,000.00
- Meal allowance during data gathering		5,000.00
- Professional fees (laboratory analysis and other related fees)		
- Miscellaneous (for incidental expenses)		5,000.00
- Printing		5,000.00
C. CAPITAL OUTLAY		
Total		P47,000.00

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(20) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

Baquete, Grayson & Mutimucio.(2016). An Exploration of Indigenous Knowledge Related to Physics Concepts Held by Senior Citizens in Chókwé, Mozambique, *International Journal of Science Education*, 38:1, 1-16, DOI: 10.1080/09500693.2015.1115137

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Moser, A., & Korstjens, I. (2018). Series: Practical guidance to qualitative research. Part 3: Sampling, data collection and analysis. *The European journal of general practice*, 24(1), 9–18. <https://doi.org/10.1080/13814788.2017.1375091>

Shaw, R., Takeuchi, Y., Uy, N., & Sharma, A. (2009). Indigenous knowledge (Disaster_risk_reduction. Policy note). European union european union, international strategy for disaster reduction, Kyoto University And Seed.

Snively, Gloria, and Corsiglia (2001). Discovering indigenous science: implications for science education. *Science Education* 85 (1):6-34.

(21) Capsule Curriculum Vitae (One-page brief C.V. only)

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